Assignment 3: Inference, models and honest limits

Statistical Thinking for Data Science • First Semester, 2026/2027

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Covering Weeks 8 to 12 • Bayes, intervals, tests, regression, Bayesian updating

How to hand this in

Work on your own. Show your working: an answer with no working earns no marks, and a correct number explained badly earns less than a wrong number reasoned carefully. Write your **name, index number and programme** at the top of the first page.
Submit a single PDF through the link announced in class, and bring the paper copy to the next session. **Due Friday 18 December 2026, 23:59.** Late work is not marked without a written reason agreed beforehand. Marked out of 50.
If you used any AI tool, add one line at the end saying which and what for. That line costs you nothing. An undeclared tool is academic dishonesty.

Part A: conditioning and intervals (20 marks)

1. A condition affects 3% of a workforce. A screening test detects 92% of real cases and flags 12% of healthy people.
   (a) Find the probability that a worker who tests positive has the condition. (3)
   (b) Show the same calculation as natural frequencies out of 10 000 workers. (3)
   (c) Write two sentences explaining the result to someone who objects that the test is 92% accurate. (2)
   (d) The occupational health officer wants to raise the proportion of positives that are genuine. Say which of sensitivity or specificity she should buy, and why. (2)

2. A sample of 40 shifts gives a mean downtime of 18.4 minutes with a standard deviation of 5.2 minutes.
   (a) Compute the standard error and a 95% confidence interval. (4)
   (b) Write one sentence stating correctly what the 95% refers to, and one stating what the interval does not cover. (3)
   (c) How many shifts would be needed to halve the width of the interval? (3)

Part B: a test and a model (20 marks)

3. Site A reports 27 incidents in 120 shifts; site B reports 18 in 130.
   (a) Compute both rates, the difference, and a 95% confidence interval for the difference. (5)
   (b) Carry out a test of the difference and report the p-value. (3)
   (c) Write the two sentence conclusion you would put in a safety report. It must mention the effect size and the uncertainty, and it must not claim the sites are equal. (4)

4. A regression of exam mark on weekly study hours, from 30 students, gives $\bar{x} = 12$ hours, $\bar{y} = 54$ marks, $s_x = 2.1$, $s_y = 7.5$ and $r = 0.84$.
   (a) Find the slope and intercept of the least squares line. (3)
   (b) Interpret the slope in the units of the problem, and state what R^2 is and what it does not tell you. (3)
   (c) Predict the mark for a student studying 15 hours, and then for one studying 40 hours. Comment on the second prediction. (2)

Part C: belief, and what you cannot say (10 marks)

5. Of 120 shifts sampled at site A, 27 had an incident. Starting from a Beta$(2, 2)$ prior, write down the posterior distribution for the incident rate and compute its mean. State the difference between the credible interval you would quote from it and the confidence interval of question 3. (5)

6. In **half a page, handwritten**, write the "limits" paragraph for the safety comparison in question 3, as if it were going to the mine manager. Be specific: name who is missing from the data, name one confounder by name, and name one decision the data cannot support. General modesty earns nothing. (5)

Total: 50 marks. Questions 1 to 4 may be typed; question 6 must be in your own handwriting.